FareShare

Verified College Ride-Pooling & Fair Distance-Split Platform

MVP MILESTONE SUBMISSION

HACKATHON EVENT	ENGINEERING TEAM	WORKING PROTOTYPE STATUS
coNStruCT 2.0 Product x AI Hackathon (NST & The Product Folks)	Team EnderMen (5 Builders)	Production Deployed (Vite 6 / React 19)

75%	< 50ms	4 Corridors	1-Click
SOLO FARE REDUCTION	ZERO-LATENCY MATCHING	AIRPORT, STATION & ISBT	DIRECT UPI SPLIT

1. What We Have Built (Production Prototype Overview)

Rather than delivering static wireframes or non-interactive mockups, **Team EnderMen** engineered and deployed a fully functional, end-to-end web platform. Users can execute complete student ride-pooling journeys in real time—from bilingual route discovery to real-time seat decrementing and pre-filled UPI settlement.

Campus-Verified Safe Circles & Filters

Features a dedicated **"Girls-Only Safe Pools"** toggle ensuring female students travel exclusively with verified female peers. Incorporates **Luggage Allowance matching** (Light Backpack, 1 Trolley, Heavy Boot Space) so passengers never arrive to an overloaded cab.

Bilingual NLP Intent Parser (Voice & Text)

Integrated with the browser **Web Speech API** and an in-browser algorithmic parser supporting English, Hindi, and Hinglish queries (e.g. *"kal shaam IGI T3 jana hai"*). Auto-extracts departure time, destination, and urgency tags without cloud latency.

Multi-Modal Solo Benchmark Strips

Every pool card features interactive comparison pills linking directly to **Uber Go**, **Ola Mini**, and the **Delhi Metro (DMRC)** official route & fare calculator, allowing users to verify prevailing solo rates with zero login barriers.

1-Click UPI Settlement & Ephemeral Privacy

In-app group ride drawers embed pre-filled **NPCI Universal UPI deep-links** (GPay, PhonePe, Paytm) carrying the exact per-person fare share. Group coordination automatically purges post-ride to protect student privacy.

2. Real-Time Economic Impact (Solo vs. FareShare 4-Way Pool)

CAMPUS TRANSIT CORRIDOR	SOLO UBER GO	SOLO OLA MINI	DELHI METRO (DMRC)	FARESHARE 4-WAY POOL	STUDENT SAVINGS
Campus ⇄ IGI Airport T3 (44 KM • ~65m)	₹1,400	₹1,350	₹60 (80m, 2 switches)	**₹350 / rider**	Save ₹1,050 (75%)
Campus ⇄ NDLS Railway Station (31 KM • ~50m)	₹900	₹880	₹50 (65m, Yellow line)	**₹225 / rider**	Save ₹675 (75%)
Campus ⇄ Anand Vihar ISBT (28 KM • ~45m)	₹800	₹780	₹50 (60m, Blue line)	**₹200 / rider**	Save ₹600 (75%)
Campus ⇄ Cyber City / Hub (14 KM • ~25m)	₹450	₹430	₹30 (40m, Rapid connector)	**₹113 / rider**	Save ₹337 (75%)

3. End-to-End User Journey (Working Live Today)

STEP 1 **Discovery & Smart Filters**

Students select a corridor or speak a Hinglish intent. They toggle **Girls-Only Safe Circles** and select their luggage profile. Live matching filters out non-compliant rides instantly.

STEP 2 **Solo Fare Cross-Check**

Before joining, students click external links to Uber Go, Ola Mini, or DMRC Metro to inspect market baseline pricing with zero sign-in friction, confirming the exact 75% savings ratio.

STEP 3 **Booking & Confirmed Pass**

Clicking **"Request to Pool"** claims an open seat in real-time state, decrements availability (preventing overbooking), and generates a verified digital Boarding Pass with peer ratings.

STEP 4 **Ride Hosting (Host Flow)**

If no matching pool exists, students click **"+ Post Ride Pool"**. The algorithmic engine computes proportional fare splits and instantly publishes the ride to the campus feed.

4. User Testing & Feedback Iterations

We validated the prototype through task-based walkthroughs with hostel and day-scholar students commuting to Delhi transit hubs. Their feedback drove five concrete engineering iterations:

Safety Feedback

Female testers cited random pooling as a blocker. Implemented dedicated **Girls-Only Safe Circles** and verified student credentials.

Luggage Friction

Airport break travel often overcrowds car boots. Added explicit **Luggage Volume Badges** to prevent mismatched cargo on arrival.

Payment Awkwardness

Testers disliked post-trip Splitwise math. Built **1-Click pre-filled UPI links** (GPay/PhonePe/Paytm) for instant on-the-spot settlement.

5. Production Roadmap: Scaling from MVP to Pan-India

The Big Vision: Moving from preset campus hubs to a universal transit mesh for every university in India.

- **Full Google Maps Platform Integration:** Dynamic pin-to-pin routing anywhere in the city using the *Google Maps Distance Matrix* and *Directions API* for algorithmic detour fare splits.
- **Pan-University Inter-College Mesh:** Expanding from single campuses to inter-college pooling networks connecting neighboring university clusters nationwide.
- **Commercial Dispatch Integration:** Official Uber & Ola Rides API webhooks for automated single-tap ride booking directly from the pool drawer.

LIVE PRODUCTION REPLIT APP
https://fareshare--pvats2112.replit.app

OPEN-SOURCE GITHUB REPOSITORY
https://github.com/Prachivats-21/fareshare